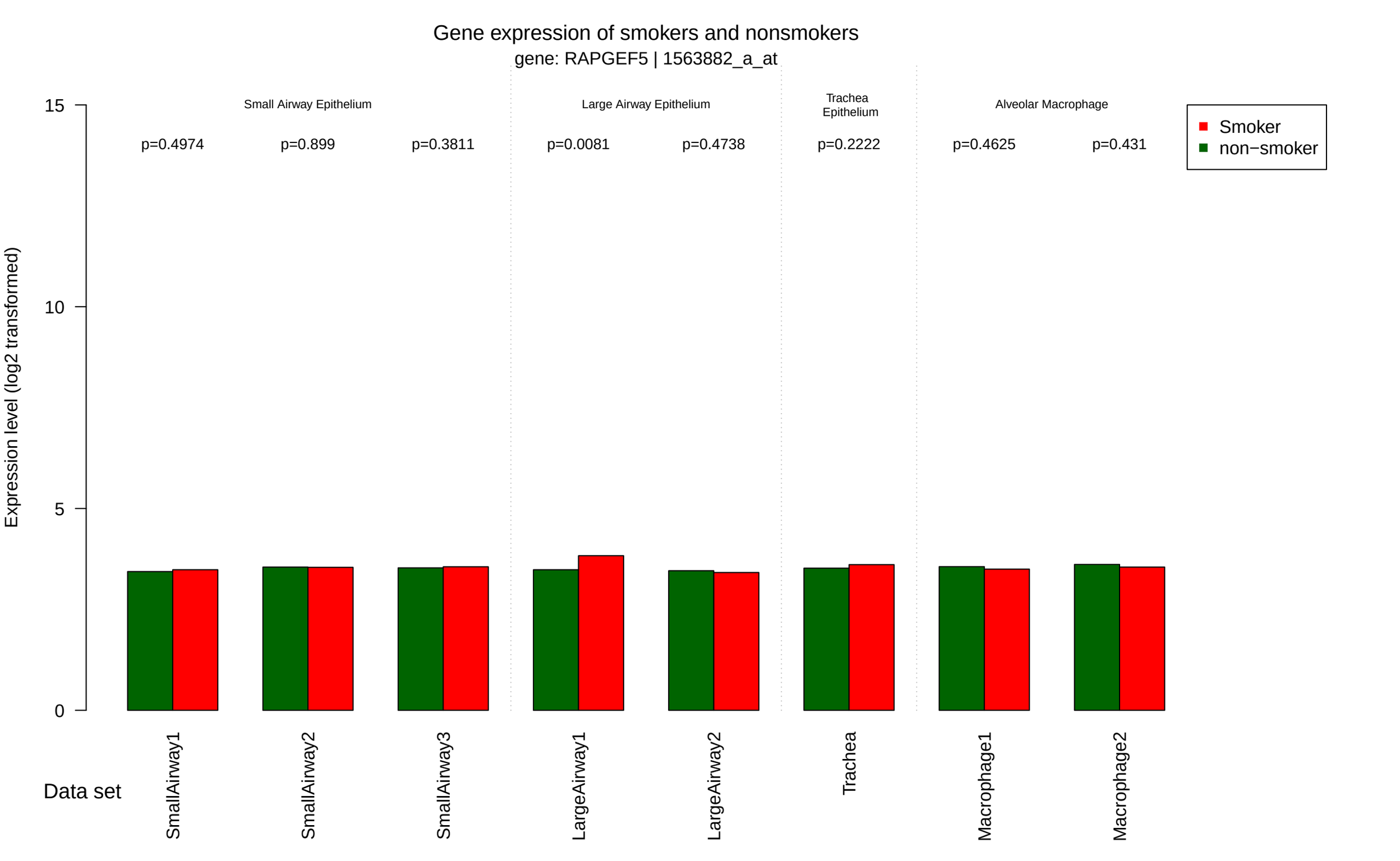
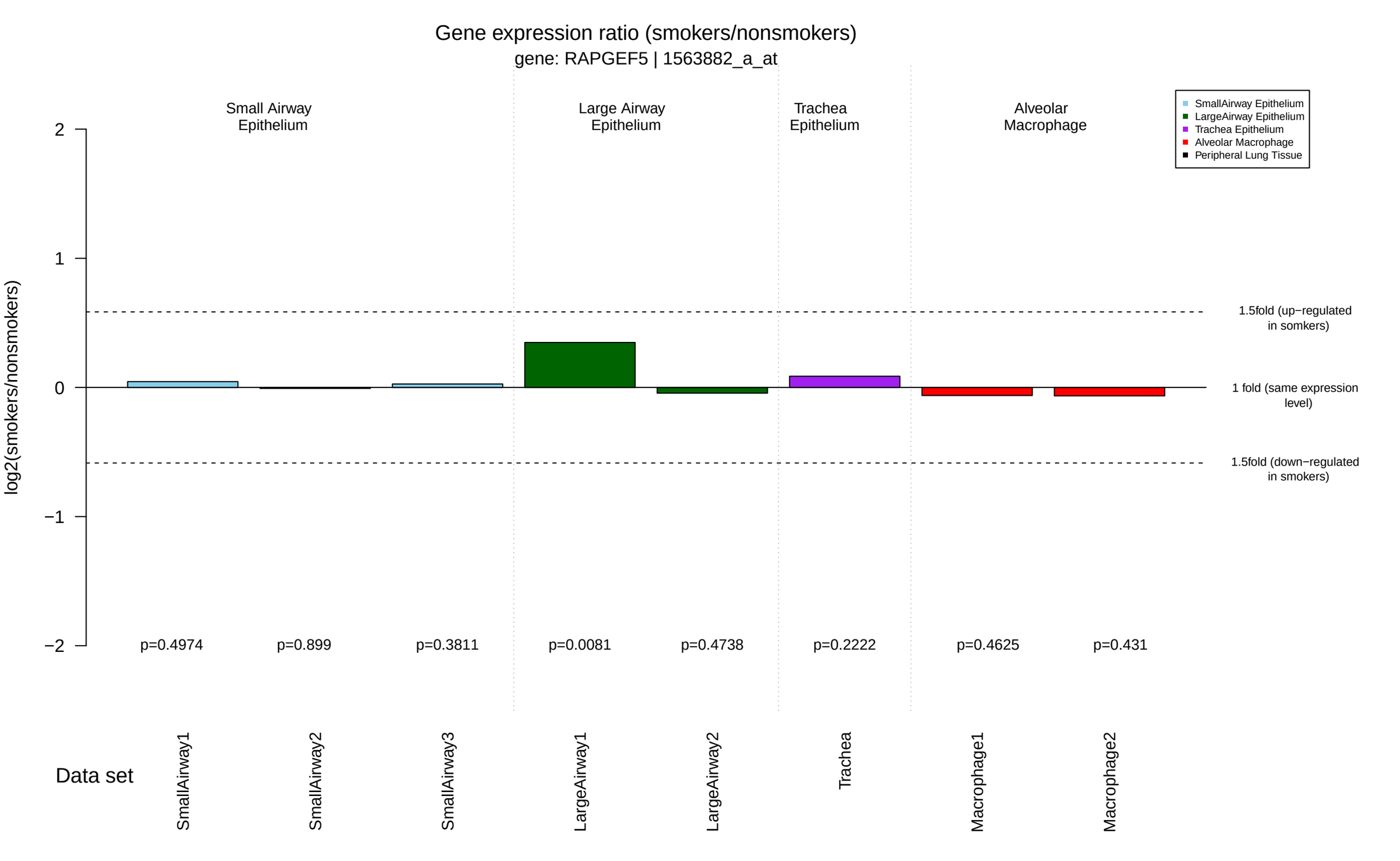
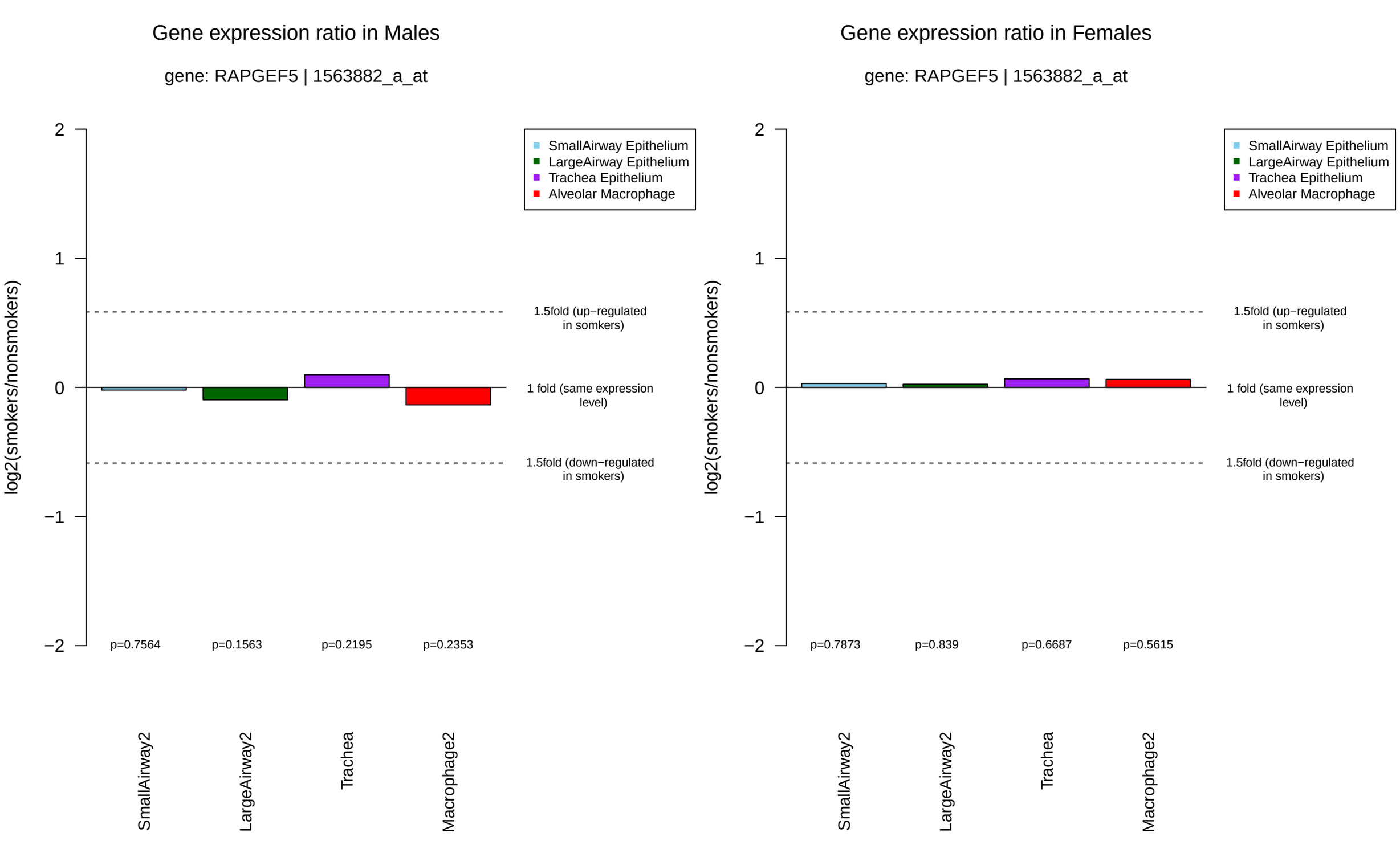
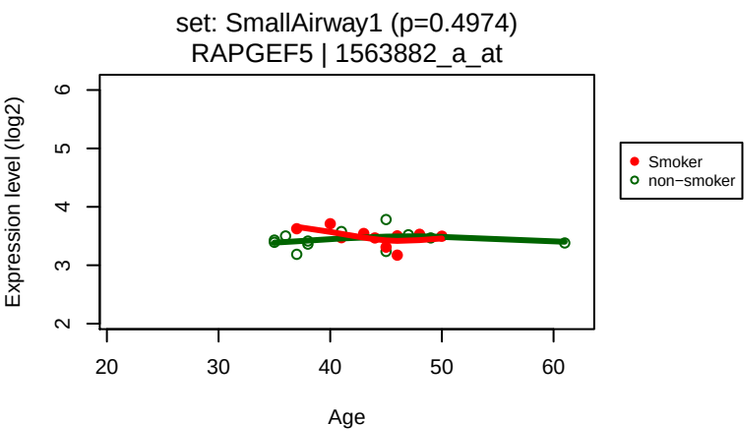
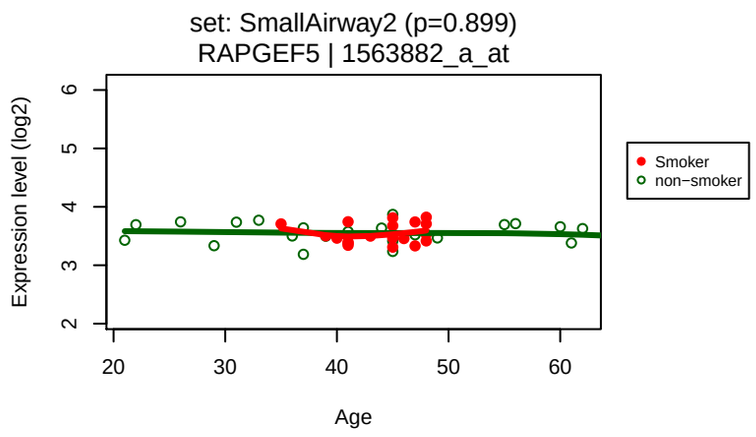
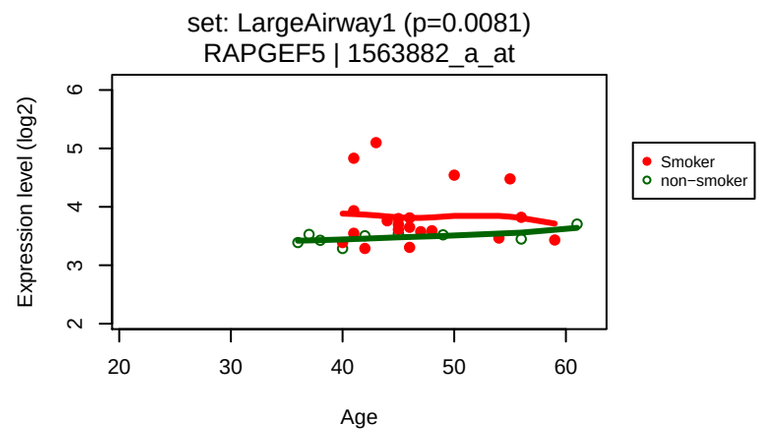

Gene expression of smokers and nonsmokers
gene: RAPGEF5 | 1563882_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4974
p=0.899
p=0.3811
p=0.0081
p=0.4738
p=0.2222
p=0.4625
p=0.431
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

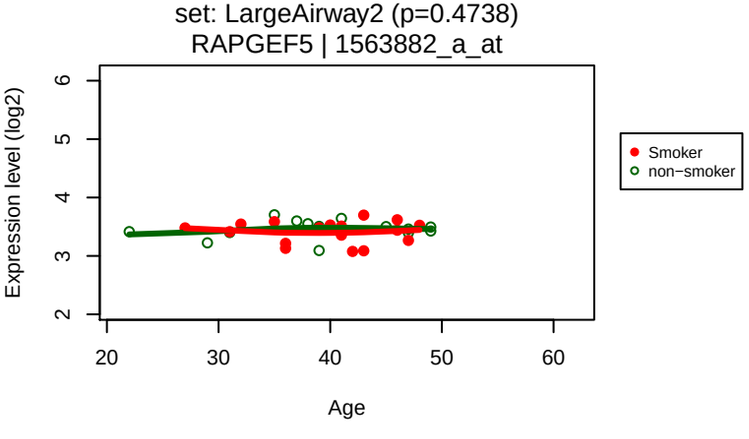

Gene expression ratio (smokers/nonsmokers)
gene: RAPGEF5 | 1563882_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4974
p=0.899
p=0.3811
p=0.0081
p=0.4738
p=0.2222
p=0.4625
p=0.431
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

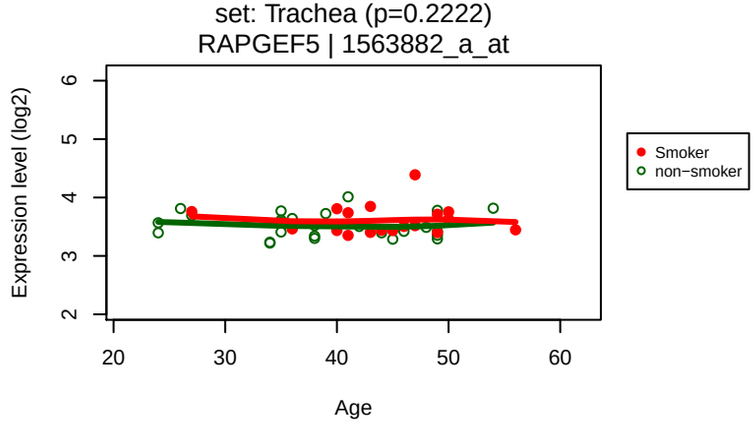

Gene expression ratio in Males
gene: RAPGEF5 | 1563882_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7564
p=0.1563
p=0.2195
p=0.2353
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAPGEF5 | 1563882_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7873
p=0.839
p=0.6687
p=0.5615
SmallAirway2
LargeAirway2
Trachea
Macrophage2

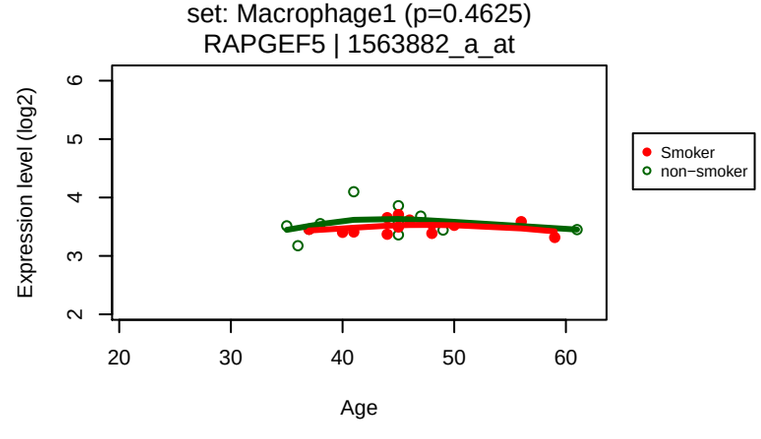

set: SmallAirway1 (p=0.4974)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker

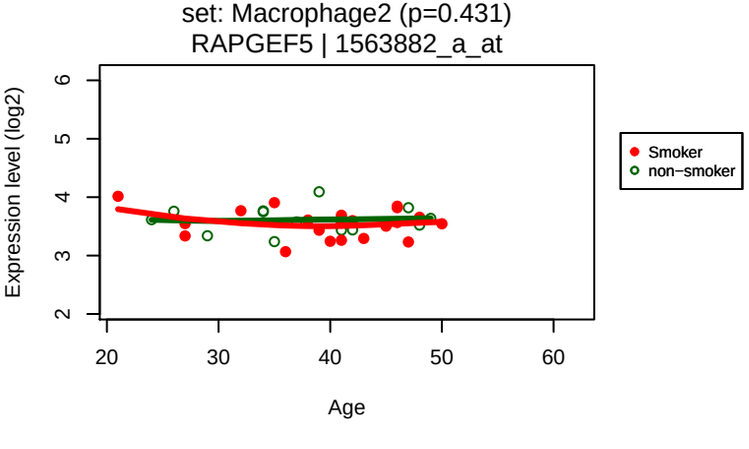

set: SmallAirway2 (p=0.899)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker

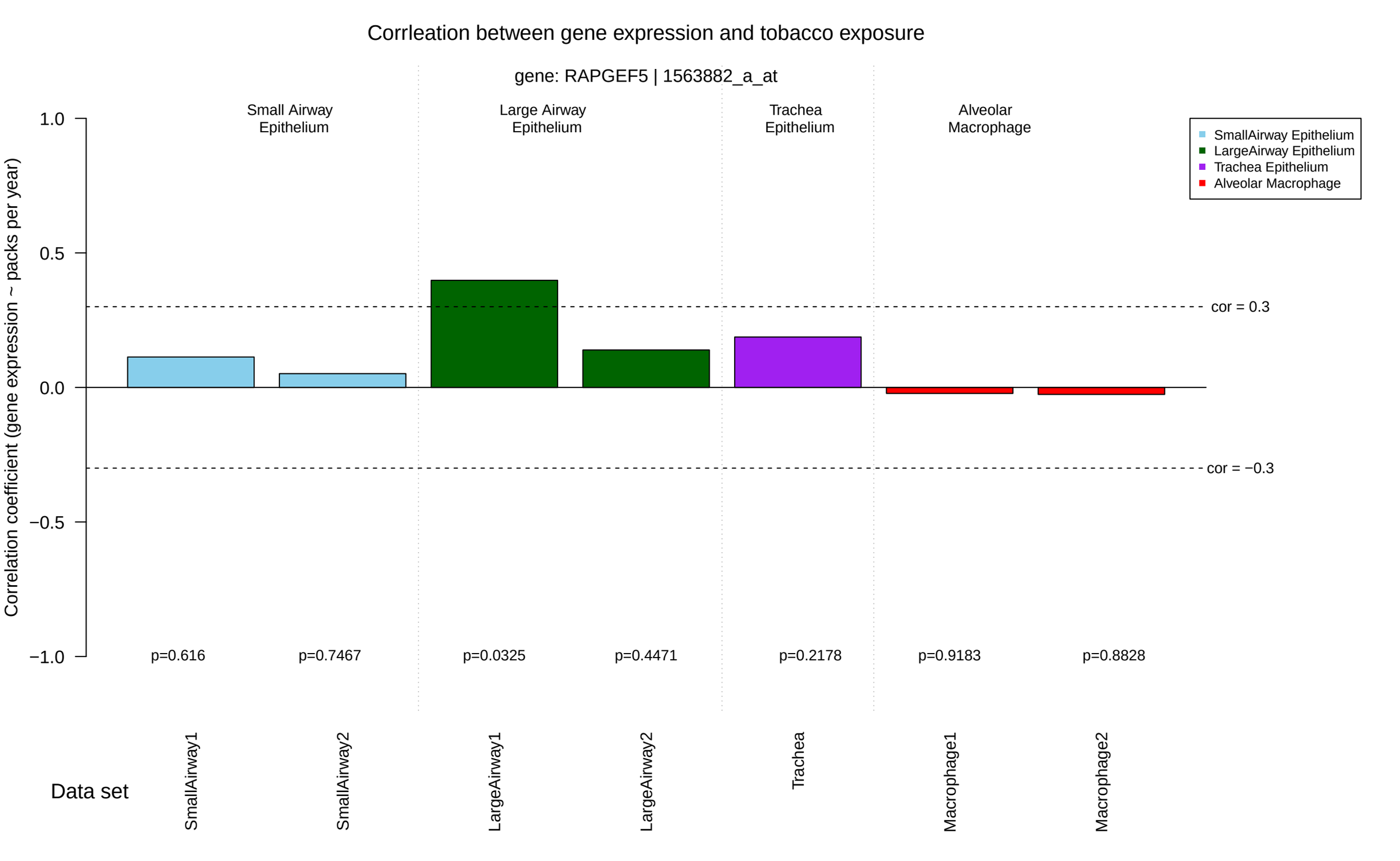

set: LargeAirway1 (p=0.0081)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4738)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2222)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4625)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.431)
RAPGEF5 | 1563882_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAPGEF5 | 1563882_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.616
p=0.7467
p=0.0325
p=0.4471
p=0.2178
p=0.9183
p=0.8828
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2